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The Underground Network We Have Been Farming Against

New 3D imaging of mycorrhizal fungi, and what it reveals about the PQNK principle of a moist, aerated, undisturbed living root zone

Asif Sharif · Pedaver · October 2026

Abstract

A 2026 New Phytologist methods paper has made it possible to visualize and quantify arbuscular mycorrhizal fungal (AMF) structures, roots, soil particles and pore space together in three dimensions within intact soil. The work does not test PQNK, rice irrigation, flooding or Alternate Wetting and Drying (AWD). Its significance for PQNK is different and potentially more important: it provides a new way to observe the physical biological architecture that PQNK has long sought to protect through no soil disturbance, no inundation, permanent organic cover and biodiversity. This Knowledge Paper separates what the new research actually demonstrates from the implications it raises for PQNK, and sets out what it means for rice grown three ways: continuously flooded, under alternate wetting and drying (AWD), and in the continuously moist but aerated PQNK root zone.

1. A hidden system becomes visible

For most of agricultural history, the most important biological infrastructure around a plant root has been largely invisible. Farmers could see the crop, roots could be excavated, and fungi could be stained or cultured, but the intact three-dimensional relationship among roots, fungal hyphae, soil particles and pore space was extraordinarily difficult to observe without destroying the very architecture being studied.

In 2026, Henri M. Braunmiller and colleagues reported a synchrotron-based X-ray micro-computed tomography workflow that can noninvasively image AMF structures in intact soil. Their method permits quantitative three-dimensional assessment of hyphal length, branching, volume and surface area, measurement of AMF-soil and AMF-root interfaces, and microscale analysis of pore-space occupancy. The authors emphasize that micro-CT can visualize AMF in air-filled soil pore space and can connect fungal architecture directly with surrounding soil structure.

Emily Gardner's Eos article, forwarded to us by Norman Uphoff, translated this methodological advance into a striking visual story. The importance is not that science has only now discovered mycorrhiza. Mycorrhizal symbiosis has been studied for generations. The advance is that parts of the network can now be reconstructed in situ, in three dimensions, with roots, fungal structures and soil retained in their spatial relationship.

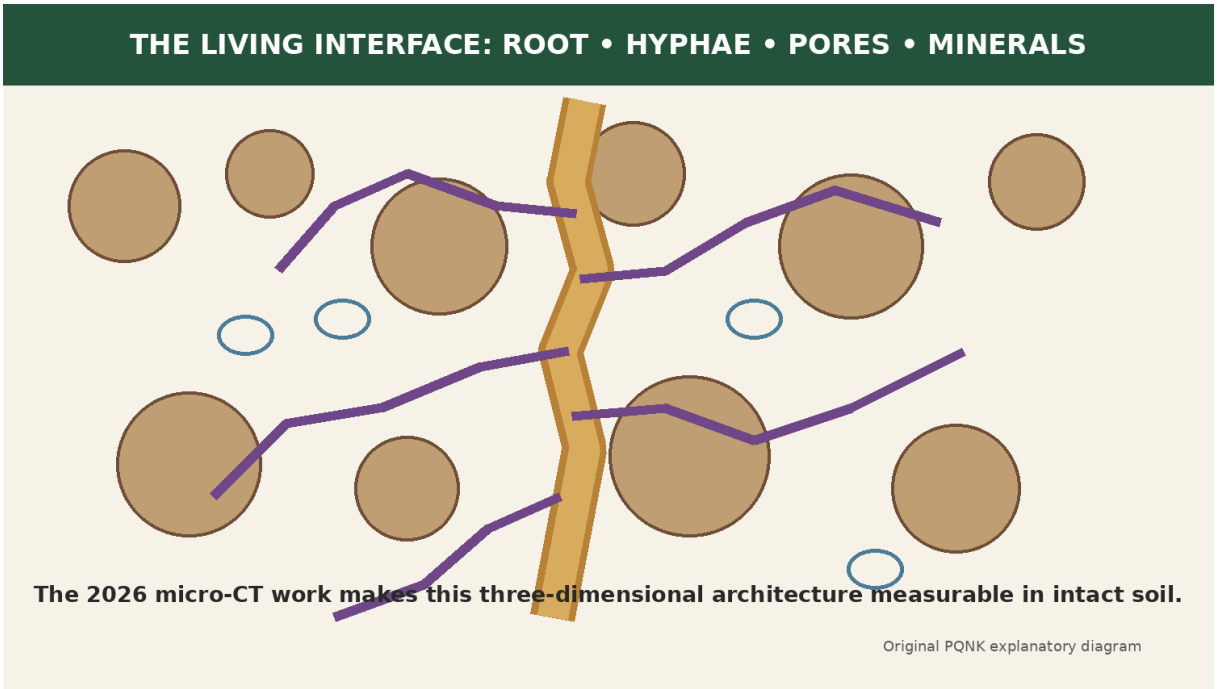


Figure 1. PQNK explanatory diagram of the root-hypha-pore-mineral interface. The diagram is interpretive, not a reproduction of the Braunmiller et al. images.

2. What Braunmiller and colleagues actually found

The study used two complementary systems and included tomato plants grown in coarse sand and silty loam with AMF. The researchers imaged extraradical fungal structures and, in rhizosphere samples, roots together with fungal structures. They demonstrated that hyphae could be detected spanning air-filled pores, forming branching networks and connecting soil particles. In rhizosphere samples, fungal networks were observed growing toward and along root surfaces and extending into surrounding soil.

The method is important because architecture matters. A fungal network is not simply a list of species. Its length, branching, connectivity, interfaces with roots and minerals, and occupation of pore space determine how the biological system is physically organized. The new workflow makes several of those features measurable without first dismantling the soil.

The authors also state an important limitation: water in fine pores can obscure fungal structures because water and fungal hyphae have similar X-ray attenuation. Their technique therefore detects hyphae most effectively in air-filled pores. This is a limitation of the imaging method, not proof that fungi cannot occur in wetter microsites.

3. Why this matters to PQNK

PQNK has consistently treated soil as a living production system rather than an inert material that must repeatedly be mechanically rebuilt and chemically supplied. Its four governing rules are designed to protect the physical and biological continuity of that system: no inundation, no soil disturbance after conversion, permanent organic cover, and encouragement of biodiversity.

The new imaging does not validate those four rules as a package, because that was not the experiment. What it does provide is direct visual and quantitative access to a part of the living architecture that those rules are intended to protect. Hyphae occupy pore space, bridge soil particles, attach to roots and extend the effective biological reach of the root system. Soil structure is therefore not merely a container for roots. It is part of the operating environment of the root-fungal partnership.

This distinction is central to PQNK. Conventional production agriculture often asks how much nutrient should be added to soil. PQNK first asks whether the biological pathways capable of finding, mobilizing, transporting and exchanging nutrients have been protected. The question is

not fertilizer versus fungi as competing products. It is whether the production system allows the natural nutrient-acquisition architecture to remain functional.

4. Air is not empty space

In production agriculture, pore air is easily treated as unused space. Biologically, it is not. Aeration governs respiration, redox conditions and the physical environment through which roots and microorganisms operate. PQNK Soil Moisture Management therefore seeks a moist but aerated root zone rather than a saturated one, at roughly 30 percent water to 70 percent air in the pore space (the 30/70 rule). Water remains essential as the carrier of dissolved minerals and as the raw material for plant physiology, but it must coexist with air in the pore system.

The Braunmiller study is especially thought-provoking because the structures that could be resolved by micro-CT were those occupying air-filled pores. That does not establish a preferred agronomic water-air ratio. It does, however, make the physical reality of air-filled biological pathways unusually visible.

PQNK's practical question is therefore not simply 'How much water did we apply?' It is 'What root-zone condition did that water create?' A field can receive water without the root zone remaining inundated. The objective is to replenish moisture while preserving the aerated biological habitat.

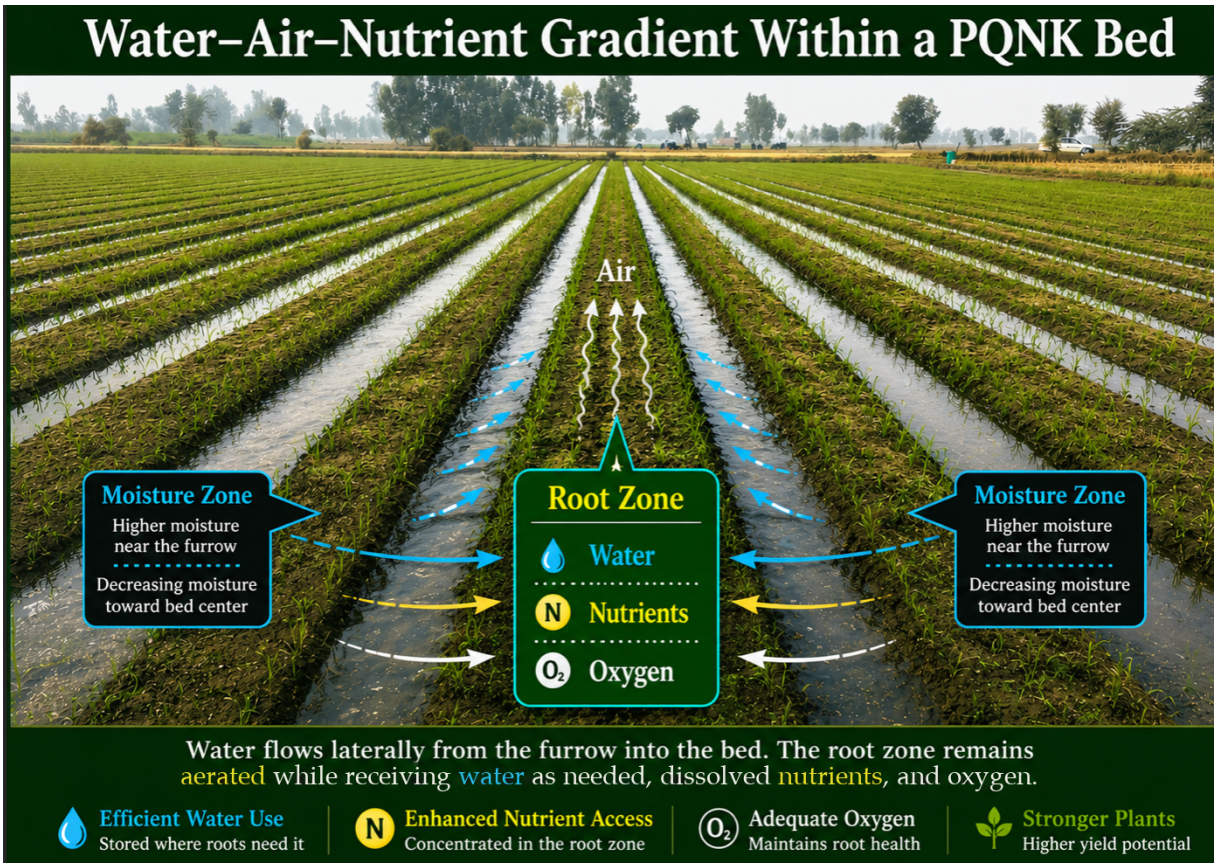


Figure 2. PQNK field-scale interpretation of water, air and nutrient movement within the permanent raised bed. Water moves laterally from the furrows while the protected production zone is managed to remain moist and aerated. The observed nutrient gradient toward the central rows is a PQNK field observation; it is not a finding of Braunmiller et al. (2026).

At field scale, water is supplied intermittently according to soil moisture need, never continuously and never above half the furrow depth. After furrow application, moisture moves laterally through the bed pore system. As it moves, water can carry dissolved nutrients while the management objective is to preserve air in the root zone. The center-row nutrient gradient shown in Figure 2, and visible in the field photographs in Figure 3, is a well-observed PQNK field condition: crop after crop, the central rows stand taller, denser and greener than the rows beside the furrows.



Figure 3. Field evidence from PQNK permanent raised beds (from Knowledge Paper KP-042): in wheat and other crops, the central rows of each bed stand taller, denser and greener than the rows beside the furrows, consistent with a better-aerated bed centre receiving a more concentrated nutrient solution. PQNK field observation.

5. Rice: flooding, AWD and the PQNK question

The email exchange that brought this article to our attention raised the question of rice. Jan Willem Ketelaar asked how rice roots behave under aerobic and anaerobic conditions, and under Alternate Wetting and Drying. The new imaging shows why the answer matters: the living architecture it reveals occupies the air-filled pores that flooding takes away.

PQNK adds a third treatment that should not be confused with AWD. Continuous flooding maintains standing water and creates a predominantly reduced bulk-soil environment. AWD deliberately alternates wetter and drier phases. PQNK seeks neither permanent flooding nor a designed cycle of flooding and drying. Its target is a continuously moist but aerated root zone, with irrigation governed by actual soil moisture and plant need.

The three regimes treat that architecture very differently. Continuous flooding fills the pores with water and starves the network of air. AWD lets air back in during each drying phase, only to drown it again at the next flooding. PQNK keeps the root zone moist and aerated throughout, so the root-fungal network is never drowned and never rebuilt: it stays in place, crop after crop.

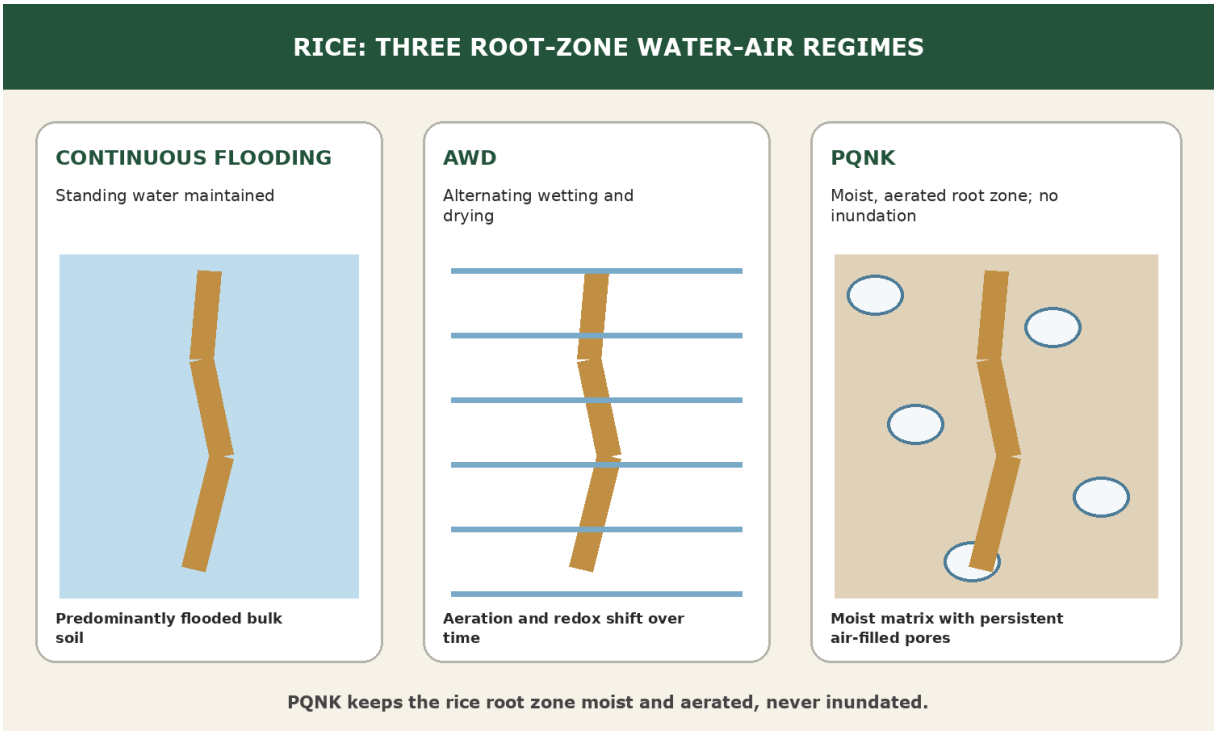


Figure 4. Three root-zone water-air regimes for rice: continuous flooding, alternate wetting and drying, and the PQNK moist, aerated root zone.

6. What PQNK has been teaching

PQNK did not invent mycorrhiza, photosynthesis, biological nitrogen fixation or natural nutrient cycling. These processes evolved in nature long before agriculture. Pedaver's work has been to identify agricultural practices that conflict with the natural production ecosystem, remove those contradictions, and develop a practical field architecture in which natural processes can perform more of the work of production.

From that perspective, tillage is not objectionable merely because fuel is consumed. Repeated disturbance breaks continuity in the soil habitat and repeatedly reconstructs the physical environment in which roots, fungi and other organisms operate. Inundation is not objectionable merely because water is used. It changes the balance between water and air in the root-zone pore system. Bare soil is not objectionable merely because it looks exposed. It loses the thermal, moisture and biological buffering supplied by cover. Reduced biodiversity matters because the soil food web functions through relationships rather than isolated organisms.

Conventional production does not disturb the soil once; it rebuilds the field after every harvest. The residue is burnt or removed, the soil is ploughed, rotavated, levelled and re-ridged, and the next crop is sown into a freshly loosened, bare seedbed. Each of these passes tears the hyphal networks, collapses the pores they occupy, separates the fungi from the roots and mineral surfaces they had connected, and exposes the soil to heat and drying. The living architecture that the new images make visible is dismantled with every crop and has to begin again from fragments, often under a fresh dose of soluble fertiliser, which is known to reduce mycorrhizal colonisation. Agriculture has been farming against this underground network not by choice but by routine. PQNK reverses the sequence: the field is engineered once, and from then on the permanent beds, retained roots, surface residues and fungal networks are carried forward from crop to crop instead of being reconstructed after every harvest.

The same holds for the System of Rice Intensification (SRI) as it is commonly practised. SRI was a major step forward: young seedlings, wide spacing and intermittent rather than continuous flooding all give roots and soil life more room and more air. But the field is still ploughed, and usually puddled, before every transplanting, and the rotary weeder stirs the topsoil several times during the season, so the underground network is still dismantled and rebuilt with each crop.

PQNK itself began in 2009 as a mechanised SRI trial on raised beds; what it added over the following years was permanence: beds that are never rebuilt, roots and residues that stay in place, and no inundation at any stage, so that the root-fungal architecture can accumulate from one crop to the next.

The new images are therefore valuable to PQNK because they help make visible what has often been described only conceptually: the crop is not operating through roots alone. It is operating through a root-associated biological and physical network embedded in the pore architecture of living soil.

7. What this research does not prove

• Braunmiller et al. did not study PQNK.

• They did not study rice.

• They did not compare flooded and non-flooded production.

• They did not compare AWD with a continuously moist, aerated root zone.

• Their ability to visualize hyphae in air-filled pores should not be interpreted as evidence that all mycorrhizal activity occurs only in dry or air-filled soil.

• The study is principally a methodological advance. It creates a new capacity to observe and quantify structures that can now be investigated under contrasting management systems.

8. What this means for rice

PQNK rice is grown without inundation. Water is given to the furrow only when the soil-ball test shows a need, never above a 4-inch (10 cm) head, so the root zone stays moist while its pores keep holding air. The root-fungal architecture that the new imaging reveals is left intact from season to season instead of being drowned with every irrigation.

Farmers already grow rice on PQNK beds without a flooded field. The new imaging makes visible what happens inside that living root zone when inundation is removed and air stays available: the network of roots and hyphae that feeds the crop remains in place.

9. The larger lesson

Industrial agriculture became extraordinarily skilled at managing what could be seen and measured above ground: seed, machinery, fertilizer, pesticide, irrigation and harvest. The biological infrastructure below ground remained comparatively opaque. That imbalance encouraged agriculture to treat soil primarily as a medium to be cultivated and supplied.

The black box is now opening. As imaging improves, the distinction between 'soil' and 'soil life' becomes increasingly artificial. Roots, fungal hyphae, microorganisms, mineral particles, organic residues, water and air form one operating environment. Management changes any one of these and therefore changes the conditions experienced by the others.

This is why PQNK's four rules must be understood together. They are not four independent techniques. They are management boundaries intended to preserve the continuity of the natural production ecosystem. The 2026 imaging work does not prove the whole PQNK system, but it gives agriculture a new window into precisely the kind of underground architecture that PQNK has been teaching farmers to protect.

PQNK Principle: Do not merely feed the plant. Protect the living architecture through which nature feeds the plant.

Image and Reuse Note

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Source Distinction

Research-derived statements in this paper are attributed to Braunmiller et al. (2026) and Gardner (2026). Statements describing PQNK principles and field architecture are Pedaver/PQNK interpretation and are not presented as findings of the Braunmiller study.